

- 4 The photograph shows mangroves in the Caribbean.



(Source: Stefania Barbier / Alamy Stock Photo)

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A new type of bacteria was discovered living in these mangroves.

Scientists called this new type of bacteria *Thiomargarita magnifica*.

One cell of these bacteria is 20 mm in length.

The smallest known type of bacteria, *Mycoplasma genitalium*, is 300 nm in length.

- (a) (i) Both of these are found in the domain Bacteria.

Give the names of the other **two** domains.

(1)

1

2

- (ii) Calculate how many times larger *T. magnifica* is than *M. genitalium*.

Give your answer in standard form.

(2)

Answer



(b) The cytoplasm of *T. magnifica* contains structures called pepins.

The function of pepins is protein synthesis.

Pepins contain circular DNA and ribosomes and are surrounded by a membrane.

Explain how these **three** structures contribute to the function of pepins.

(3)

DNA

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Ribosomes

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Membrane

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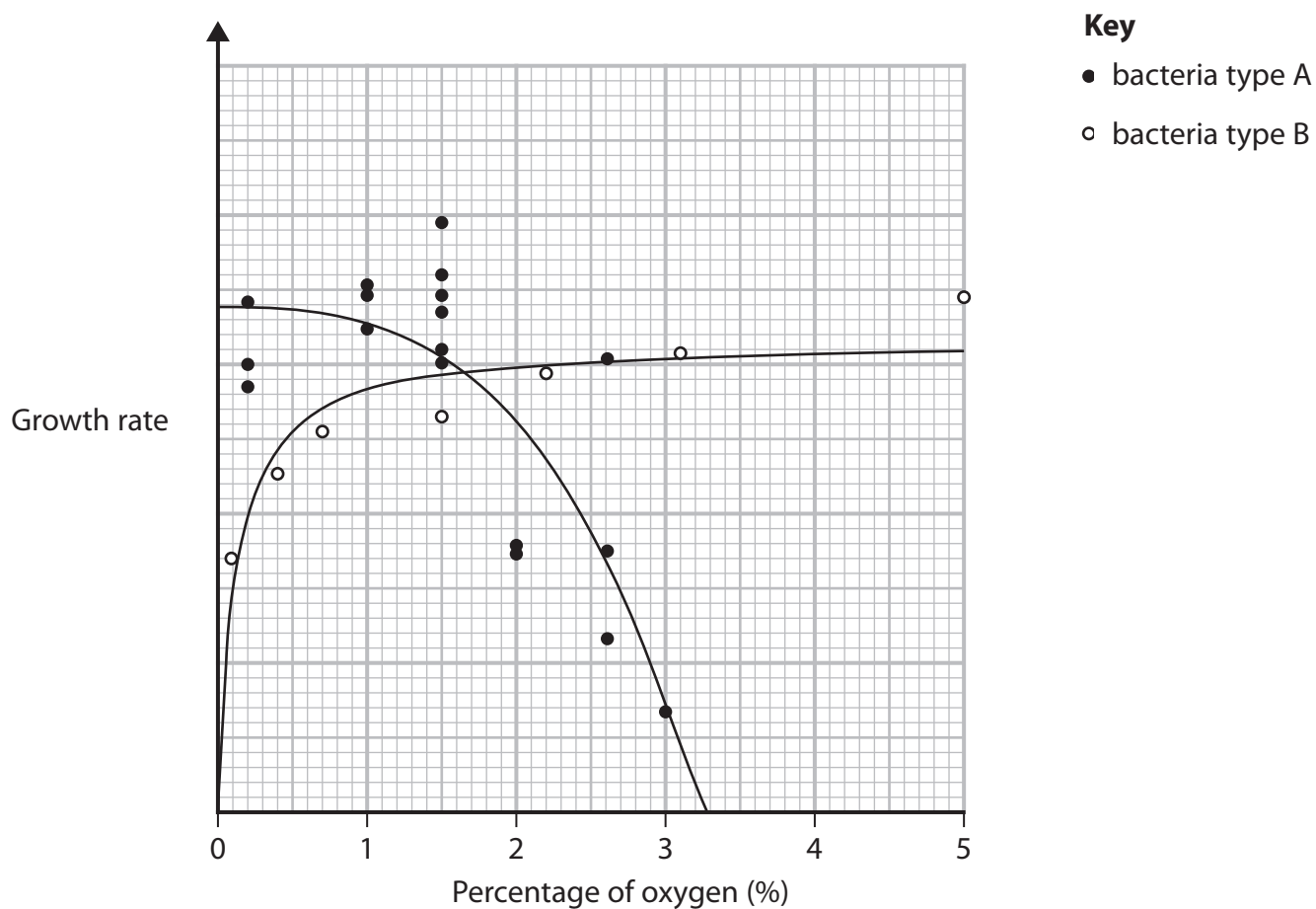
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- (c) Some bacteria require oxygen in their surroundings whereas other bacteria do not.

The graph shows the growth rate of two types of bacteria, A and B, which were grown in different concentrations of oxygen.

The bacteria were supplied with water and grown at their optimum temperatures.



- (i) Explain why an optimum temperature and water are needed for bacterial growth.

(2)

Optimum temperature

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Water

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- (ii) Explain the results for these two types of bacteria shown in the graph.

(3)

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(Total for Question 4 = 11 marks)

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